Generowanie danych

Uwaga prez pomyłke nie dałem set seed!

```
set.seed(122112)
n<-300
p<-20
X<-matrix(rnorm(n*p,2,5),ncol=p,nrow=n)
X[,1]<-rep(1,n)
beta<-c(rnorm(10,0,1/4),numeric(10))
pi.p<-1/(1+exp(-X%*%beta))
colnames(X)<-sapply(1:p,function(i){paste0("X",i)})
dane<-data.frame(X)
dane$Y<-rbinom(n,1,pi.p)
write.csv(dane,"daneG5.csv",row.names=F)
```

```
id.0<-which(dane$Y==0)
id.1<-which(dane$Y==1)

train.0<-sample(id.0,size=floor(0.7*length(id.0)))
train.1<-sample(id.1,size=floor(0.7*length(id.1)))

id.train<-sort(c(train.0,train.1))
id.test<-setdiff(seq_len(n),id.train)

train<-dane[id.train,]
test<-dane[id.test,]
```

Konstrukcja modelu2

```
model.full<-glm(Y~.,data=train[,-1],family=binomial)
model.null<-glm(Y~1,data=train[,-1],family=binomial)

model2 <- glm(Y ~ ., data = train[, -1], family = binomial)
```

```
pred.test<-predict(model2,
                   newdata=test,type='response')

wskaźniki<-function(poziom=0.5){
  pred.class<-ifelse(pred.test>poziom,1,0)
  cm<-table(pred=pred.class,real=test$Y)
  sen<-cm[2,2]/sum(cm[,2])
  spec<-cm[1,1]/sum(cm[,1])
  fpr<-cm[2,1]/sum(cm[,1])
  fnr<-cm[1,2]/sum(cm[,2])
  acc<-sum(diag(cm))/sum(cm)
  miary<-c(sen,spec,fpr,fnr,acc)
  names(miary)<-c("sen","spec","fpr","fnr","acc")
  return(list(cm=cm,miary=miary))
}

M.1<-wskaźniki(0.5)
```

```
M.2<-wskaźniki(0.7)

cbind(M.1$miary,M.2$miary)

##            [,1]       [,2]
## sen  0.52631579 0.47368421
## spec 0.98611111 0.98611111
## fpr  0.01388889 0.01388889
## fnr  0.47368421 0.52631579
## acc  0.89010989 0.87912088
```

```
library(pROC)
roc.obj<-roc(test2$Y,pred.test2)

## Error: object 'test2' not found

auc(roc.obj)

## Error: object 'roc.obj' not found

plot(roc.obj,col='red')

## Error: object 'roc.obj' not found
```

Konstrukcja modelu

```
model.full<-glm(Y~.,data=train[,-1],family=binomial(link="probit"))
model.null<-glm(Y~1,data=train[,-1],family=binomial(link="probit"))

model <- step(model.null, scope = list(lower = model.null, upper = model.full), direction = "forward")

## Start:  AIC=214.45
## Y ~ 1
##
##        Df Deviance    AIC
## + X8    1   187.85 191.85
## + X9    1   196.05 200.05
## + X3    1   201.27 205.27
## + X6    1   202.90 206.90
## + X2    1   205.15 209.15
## + X4    1   205.26 209.26
## + X12   1   206.08 210.08
## + X10   1   206.59 210.59
## + X19   1   208.56 212.56
## + X5    1   209.04 213.04
## + X7    1   209.37 213.37
## + X18   1   209.98 213.98
## + X15   1   210.10 214.10
## <none>      212.45 214.45
## + X14   1   210.89 214.89
## + X20   1   211.59 215.59
## + X11   1   212.31 216.31
## + X17   1   212.32 216.32
## + X16   1   212.45 216.45
## + X13   1   212.45 216.45
```

```
##
## Step:  AIC=191.85
## Y ~ X8
##
##        Df Deviance    AIC
## + X9    1   173.19 179.19
## + X6    1   175.77 181.77
## + X10   1   177.93 183.93
## + X4    1   178.64 184.64
## + X3    1   178.66 184.66
## + X5    1   180.03 186.03
## + X12   1   181.29 187.29
## + X2    1   183.04 189.04
## + X7    1   184.83 190.83
## + X18   1   184.88 190.88
## + X19   1   185.40 191.40
## <none>      187.85 191.85
## + X14   1   186.59 192.59
## + X15   1   186.65 192.65
## + X11   1   187.67 193.67
## + X20   1   187.76 193.76
## + X17   1   187.82 193.82
## + X13   1   187.83 193.83
## + X16   1   187.83 193.83
##
## Step:  AIC=179.19
## Y ~ X8 + X9
##
##        Df Deviance    AIC
## + X6    1   156.98 164.98
## + X4    1   160.54 168.54
## + X3    1   161.34 169.34
## + X10   1   161.70 169.70
## + X5    1   164.92 172.92
## + X12   1   167.12 175.12
## + X7    1   169.46 177.46
## + X2    1   170.81 178.81
## + X18   1   170.94 178.94
## <none>      173.19 179.19
## + X19   1   171.28 179.28
## + X14   1   171.71 179.71
## + X15   1   172.49 180.49
## + X13   1   172.58 180.58
## + X20   1   172.98 180.98
## + X16   1   173.19 181.19
## + X11   1   173.19 181.19
## + X17   1   173.19 181.19
##
## Step:  AIC=164.98
## Y ~ X8 + X9 + X6
##
##        Df Deviance    AIC
## + X4    1   142.08 152.08
## + X3    1   146.54 156.54
## + X10   1   148.31 158.31
## + X5    1   149.78 159.78
```

```
## + X12   1   152.65 162.65
## + X7    1   152.87 162.87
## + X19   1   153.42 163.42
## + X18   1   154.53 164.53
## + X2    1   154.57 164.57
## <none>      156.98 164.98
## + X15   1   155.42 165.42
## + X14   1   155.83 165.83
## + X11   1   156.37 166.37
## + X20   1   156.64 166.64
## + X13   1   156.72 166.72
## + X17   1   156.74 166.74
## + X16   1   156.75 166.75
##
## Step:  AIC=152.08
## Y ~ X8 + X9 + X6 + X4
##
##        Df Deviance    AIC
## + X3    1   126.34 138.34
## + X7    1   133.68 145.68
## + X10   1   134.27 146.27
## + X19   1   137.21 149.21
## + X5    1   137.84 149.84
## + X12   1   139.77 151.77
## + X2    1   140.00 152.00
## + X14   1   140.06 152.06
## <none>      142.08 152.08
## + X20   1   141.02 153.02
## + X18   1   141.09 153.09
## + X17   1   141.27 153.27
## + X16   1   141.74 153.74
## + X15   1   141.79 153.79
## + X11   1   141.91 153.91
## + X13   1   142.04 154.04
##
## Step:  AIC=138.34
## Y ~ X8 + X9 + X6 + X4 + X3
##
##        Df Deviance    AIC
## + X10   1   111.86 125.86
## + X7    1   113.18 127.18
## + X5    1   116.41 130.41
## + X19   1   119.08 133.08
## + X12   1   122.48 136.49
## + X14   1   124.15 138.15
## <none>      126.34 138.34
## + X18   1   124.66 138.66
## + X17   1   124.89 138.89
## + X16   1   125.00 139.00
## + X20   1   125.70 139.70
## + X2    1   125.73 139.73
## + X15   1   125.87 139.87
## + X13   1   126.31 140.31
## + X11   1   126.34 140.34
##
## Step:  AIC=125.86
```

```
## Y ~ X8 + X9 + X6 + X4 + X3 + X10
##
##        Df Deviance    AIC
## + X7    1   98.533 114.53
## + X5    1   99.983 115.98
## + X19   1  104.251 120.25
## + X12   1  107.291 123.29
## + X14   1  109.830 125.83
## <none>     111.859 125.86
## + X16   1  109.939 125.94
## + X17   1  110.233 126.23
## + X2    1  110.575 126.58
## + X18   1  110.862 126.86
## + X15   1  111.118 127.12
## + X20   1  111.345 127.34
## + X11   1  111.739 127.74
## + X13   1  111.819 127.82
##
## Step:  AIC=114.53
## Y ~ X8 + X9 + X6 + X4 + X3 + X10 + X7
##        Df Deviance    AIC
## + X5    1   82.851 100.85
## + X19   1   92.743 110.74
## + X16   1   95.488 113.49
## + X17   1   96.035 114.03
## + X12   1   96.227 114.23
## <none>      98.533 114.53
## + X18   1   97.334 115.33
## + X14   1   97.832 115.83
## + X13   1   98.149 116.15
## + X2    1   98.221 116.22
## + X15   1   98.376 116.38
## + X11   1   98.463 116.46
## + X20   1   98.485 116.48
##
## Step:  AIC=100.85
## Y ~ X8 + X9 + X6 + X4 + X3 + X10 + X7 + X5
##        Df Deviance     AIC
## + X16   1   78.013  98.013
## + X19   1   79.978  99.978
## <none>      82.851 100.851
## + X17   1   80.880 100.880
## + X12   1   81.596 101.596
## + X18   1   82.275 102.275
## + X20   1   82.380 102.380
## + X13   1   82.553 102.553
## + X14   1   82.595 102.595
## + X15   1   82.716 102.716
## + X2    1   82.841 102.841
## + X11   1   82.845 102.845
##
## Step:  AIC=98.01
## Y ~ X8 + X9 + X6 + X4 + X3 + X10 + X7 + X5 + X16
##        Df Deviance     AIC
## + X12   1   75.020  97.020
## + X19   1   75.796  97.796
```

```
## <none>      78.013  98.013
## + X17   1   77.064  99.064
## + X14   1   77.090  99.090
## + X15   1   77.700  99.700
## + X20   1   77.712  99.712
## + X18   1   77.847  99.847
## + X13   1   77.936  99.936
## + X11   1   77.972  99.972
## + X2    1   78.000 100.000
##
## Step:  AIC=97.02
## Y ~ X8 + X9 + X6 + X4 + X3 + X10 + X7 + X5 + X16 + X12
##        Df Deviance    AIC
## <none>      75.020 97.020
## + X19   1   73.169 97.169
## + X14   1   74.308 98.308
## + X20   1   74.477 98.477
## + X13   1   74.563 98.563
## + X15   1   74.658 98.658
## + X17   1   74.699 98.699
## + X18   1   74.858 98.858
## + X2    1   75.013 99.013
## + X11   1   75.019 99.019
```

```
pred.test<-predict(model,
                   newdata=test,type='response')

wskaźniki<-function(poziom=0.5){
  pred.class<-ifelse(pred.test>poziom,1,0)
  cm<-table(pred=pred.class,real=test$Y)
  sen<-cm[2,2]/sum(cm[,2])
  spec<-cm[1,1]/sum(cm[,1])
  fpr<-cm[2,1]/sum(cm[,1])
  fnr<-cm[1,2]/sum(cm[,2])
  acc<-sum(diag(cm))/sum(cm)
  miary<-c(sen,spec,fpr,fnr,acc)
  names(miary)<-c("sen","spec","fpr","fnr","acc")
  return(list(cm=cm,miary=miary))
}

Mm.1<-wskaźniki(0.5)
Mm.2<-wskaźniki(0.7)

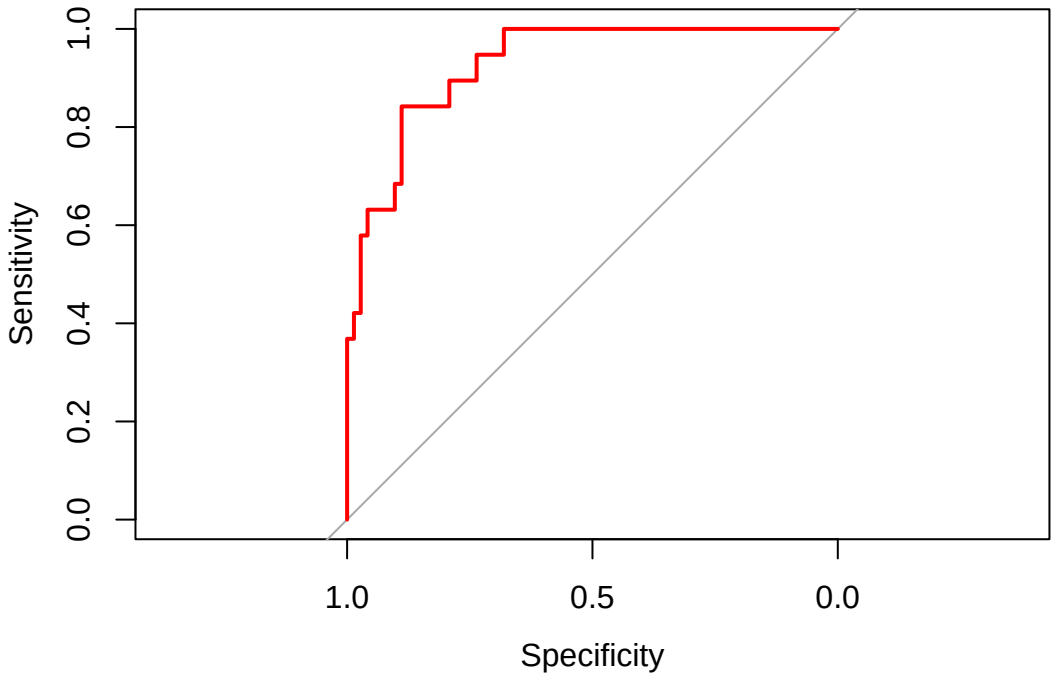
cbind(M.1$miary,M.2$miary)
```

```
##            [,1]       [,2]
## sen  0.52631579 0.47368421
## spec 0.98611111 0.98611111
## fpr  0.01388889 0.01388889
## fnr  0.47368421 0.52631579
## acc  0.89010989 0.87912088
```

```
library(pROC)
roc.obj<-roc(test$Y,pred.test)
auc(roc.obj)
```

```
## Area under the curve: 0.9284

plot(roc.obj,col='red')
```



Konstrukcja modelu3

```
model.full<-glm(Y~.,data=train[,-1],family=binomial(link="probit"))
model.null<-glm(Y~1,data=train[,-1],family=binomial(link="probit"))

model3 <- step(model.null, scope = list(lower = model.null, upper = model.full), direction = "forward",  k

## Start:  AIC=217.79
## Y ~ 1
##
##        Df Deviance    AIC
## + X8    1   187.85 198.53
## + X9    1   196.05 206.74
## + X3    1   201.27 211.96
## + X6    1   202.90 213.59
## + X2    1   205.15 215.84
## + X4    1   205.26 215.95
## + X12   1   206.08 216.76
## + X10   1   206.59 217.28
## <none>      212.45 217.79
## + X19   1   208.56 219.24
## + X5    1   209.04 219.73
## + X7    1   209.37 220.06
## + X18   1   209.98 220.66
## + X15   1   210.10 220.78
## + X14   1   210.89 221.57
```

```
## + X20   1   211.59 222.27
## + X11   1   212.31 223.00
## + X17   1   212.32 223.01
## + X16   1   212.45 223.13
## + X13   1   212.45 223.13
##
## Step:  AIC=198.53
## Y ~ X8
##
##        Df Deviance    AIC
## + X9    1   173.19 189.21
## + X6    1   175.77 191.80
## + X10   1   177.93 193.96
## + X4    1   178.64 194.67
## + X3    1   178.66 194.69
## + X5    1   180.03 196.06
## + X12   1   181.29 197.32
## <none>      187.85 198.53
## + X2    1   183.04 199.07
## + X7    1   184.83 200.85
## + X18   1   184.88 200.91
## + X19   1   185.40 201.43
## + X14   1   186.59 202.62
## + X15   1   186.65 202.68
## + X11   1   187.67 203.69
## + X20   1   187.76 203.78
## + X17   1   187.82 203.84
## + X13   1   187.83 203.86
## + X16   1   187.83 203.86
##
## Step:  AIC=189.21
## Y ~ X8 + X9
##
##        Df Deviance    AIC
## + X6    1   156.98 178.35
## + X4    1   160.54 181.91
## + X3    1   161.34 182.71
## + X10   1   161.70 183.07
## + X5    1   164.92 186.29
## + X12   1   167.12 188.49
## <none>      173.19 189.21
## + X7    1   169.46 190.83
## + X2    1   170.81 192.18
## + X18   1   170.94 192.31
## + X19   1   171.28 192.65
## + X14   1   171.71 193.08
## + X15   1   172.49 193.86
## + X13   1   172.58 193.95
## + X20   1   172.98 194.35
## + X16   1   173.19 194.56
## + X11   1   173.19 194.56
## + X17   1   173.19 194.56
##
## Step:  AIC=178.35
## Y ~ X8 + X9 + X6
##
```

```
##        Df Deviance    AIC
## + X4    1   142.08 168.79
## + X3    1   146.54 173.25
## + X10   1   148.31 175.02
## + X5    1   149.78 176.49
## <none>      156.98 178.35
## + X12   1   152.65 179.36
## + X7    1   152.87 179.58
## + X19   1   153.42 180.13
## + X18   1   154.53 181.24
## + X2    1   154.57 181.28
## + X15   1   155.42 182.13
## + X14   1   155.83 182.54
## + X11   1   156.37 183.08
## + X20   1   156.64 183.35
## + X13   1   156.72 183.43
## + X17   1   156.74 183.46
## + X16   1   156.75 183.46
##
## Step:  AIC=168.79
## Y ~ X8 + X9 + X6 + X4
##
##        Df Deviance    AIC
## + X3    1   126.34 158.40
## + X7    1   133.68 165.73
## + X10   1   134.27 166.32
## <none>      142.08 168.79
## + X19   1   137.21 169.26
## + X5    1   137.84 169.89
## + X12   1   139.77 171.82
## + X2    1   140.00 172.06
## + X14   1   140.06 172.11
## + X20   1   141.02 173.07
## + X18   1   141.09 173.15
## + X17   1   141.27 173.33
## + X16   1   141.74 173.80
## + X15   1   141.79 173.84
## + X11   1   141.91 173.97
## + X13   1   142.04 174.09
##
## Step:  AIC=158.4
## Y ~ X8 + X9 + X6 + X4 + X3
##
##        Df Deviance    AIC
## + X10   1   111.86 149.26
## + X7    1   113.18 150.57
## + X5    1   116.41 153.81
## + X19   1   119.08 156.48
## <none>      126.34 158.40
## + X12   1   122.48 159.88
## + X14   1   124.15 161.54
## + X18   1   124.66 162.06
## + X17   1   124.89 162.28
## + X16   1   125.00 162.40
## + X20   1   125.70 163.10
## + X2    1   125.73 163.13
```

```
## + X15   1   125.87 163.26
## + X13   1   126.31 163.71
## + X11   1   126.34 163.74
##
## Step:  AIC=149.26
## Y ~ X8 + X9 + X6 + X4 + X3 + X10
##
##        Df Deviance    AIC
## + X7    1   98.533 141.27
## + X5    1   99.983 142.72
## + X19   1  104.251 146.99
## <none>     111.859 149.26
## + X12   1  107.291 150.03
## + X14   1  109.830 152.57
## + X16   1  109.939 152.68
## + X17   1  110.233 152.97
## + X2    1  110.575 153.31
## + X18   1  110.862 153.60
## + X15   1  111.118 153.86
## + X20   1  111.345 154.08
## + X11   1  111.739 154.48
## + X13   1  111.819 154.56
##
## Step:  AIC=141.27
## Y ~ X8 + X9 + X6 + X4 + X3 + X10 + X7
##        Df Deviance    AIC
## + X5    1   82.851 130.93
## + X19   1   92.743 140.82
## <none>      98.533 141.27
## + X16   1   95.488 143.57
## + X17   1   96.035 144.12
## + X12   1   96.227 144.31
## + X18   1   97.334 145.41
## + X14   1   97.832 145.91
## + X13   1   98.149 146.23
## + X2    1   98.221 146.30
## + X15   1   98.376 146.46
## + X11   1   98.463 146.54
## + X20   1   98.485 146.57
##
## Step:  AIC=130.93
## Y ~ X8 + X9 + X6 + X4 + X3 + X10 + X7 + X5
##        Df Deviance    AIC
## <none>      82.851 130.93
## + X16   1   78.013 131.44
## + X19   1   79.978 133.40
## + X17   1   80.880 134.30
## + X12   1   81.596 135.02
## + X18   1   82.275 135.70
## + X20   1   82.380 135.80
## + X13   1   82.553 135.98
## + X14   1   82.595 136.02
## + X15   1   82.716 136.14
## + X2    1   82.841 136.26
## + X11   1   82.845 136.27
```

```r
pred.test<-predict(model3,
                   newdata=test,type='response')

wskaźniki<-function(poziom=0.5){
  pred.class<-ifelse(pred.test>poziom,1,0)
  cm<-table(pred=pred.class,real=test$Y)
  sen<-cm[2,2]/sum(cm[,2])
  spec<-cm[1,1]/sum(cm[,1])
  fpr<-cm[2,1]/sum(cm[,1])
  fnr<-cm[1,2]/sum(cm[,2])
  acc<-sum(diag(cm))/sum(cm)
  miary<-c(sen,spec,fpr,fnr,acc)
  names(miary)<-c("sen","spec","fpr","fnr","acc")
  return(list(cm=cm,miary=miary))
}

Mmm.1<-wskaźniki(0.5)
Mmm.2<-wskaźniki(0.7)

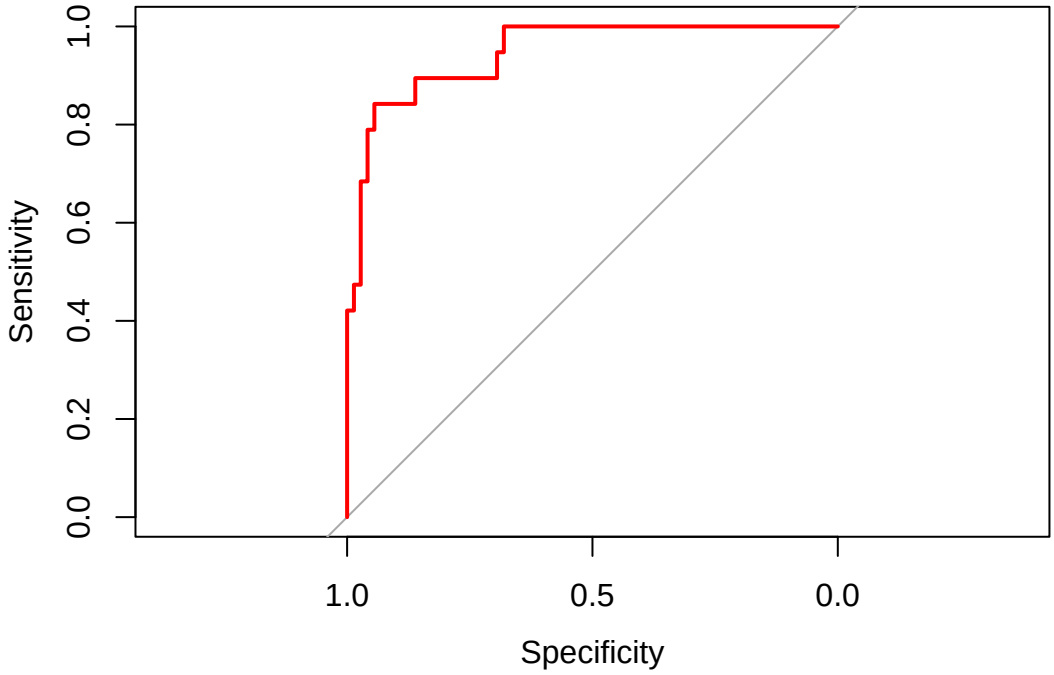
cbind(Mmm.1$miary,Mmm.2$miary)
```

```
##            [,1]       [,2]
## sen  0.52631579 0.42105263
## spec 0.97222222 0.98611111
## fpr  0.02777778 0.01388889
## fnr  0.47368421 0.57894737
## acc  0.87912088 0.86813187
```

```r
library(pROC)
roc.obj<-roc(test$Y,pred.test)
auc(roc.obj)
```

```
## Area under the curve: 0.9459
```

```r
plot(roc.obj,col='red')
```




```
nazwy <- names(coef(model))
nazwy2 <- names(coef(model2))
nazwy3 <- names(coef(model3))
wynik <- cbind(beta=beta, oszacowanieAIC = coef(model)[nazwy], oszacowanielog = coef(model2)[nazwy2], osza
wynik[is.na(wynik)] <- 0
wynik
```

```
##                    beta oszacowanieAIC oszacowanielog oszacowanieBIC
## (Intercept)  0.14965846    -0.36704727    -0.70660281    -0.03218536
## X2           0.06417778    -0.31469333     0.04927675    -0.28099242
## X3          -0.38625137    -0.24911333    -0.55790513    -0.23134936
## X4          -0.35757672    -0.08973378    -0.53397016    -0.08944111
## X5          -0.25303450    -0.25637787    -0.24594486    -0.23943460
## X6          -0.05132656    -0.27939895    -0.15898244    -0.23943757
## X7          -0.19396936     0.19086969    -0.27831466     0.16728488
## X8          -0.43647424    -0.12965390    -0.61893334    -0.13141363
## X9          -0.27908554    -0.14311099    -0.49303229    -0.12971796
## X10          0.18499581     0.08274142     0.33642440    -0.03218536
## X11          0.00000000     0.05850419    -0.01073633    -0.28099242
## X12          0.00000000    -0.36704727     0.11740196    -0.23134936
## X13          0.00000000    -0.31469333    -0.06232212    -0.08944111
## X14          0.00000000    -0.24911333    -0.03938422    -0.23943460
## X15          0.00000000    -0.08973378     0.05943756    -0.23943757
## X16          0.00000000    -0.25637787     0.14454710     0.16728488
## X17          0.00000000    -0.27939895     0.05090128    -0.13141363
## X18          0.00000000     0.19086969    -0.05201295    -0.12971796
## X19          0.00000000    -0.12965390     0.08932630    -0.03218536
## X20          0.00000000    -0.14311099     0.07439039    -0.28099242
```

```
cbind(log05=M.1$miary,log07=M.2$miary, AIC05=Mm.1$miary,AIC07=Mm.2$miary, BIC05=Mmm.1$miary,BIC07=Mmm.2$mi

##           log05      log07      AIC05      AIC07      BIC05      BIC07
## sen  0.52631579 0.47368421 0.57894737 0.42105263 0.52631579 0.42105263
## spec 0.98611111 0.98611111 0.97222222 0.97222222 0.97222222 0.98611111
## fpr  0.01388889 0.01388889 0.02777778 0.02777778 0.02777778 0.01388889
## fnr  0.47368421 0.52631579 0.42105263 0.57894737 0.47368421 0.57894737
## acc  0.89010989 0.87912088 0.89010989 0.85714286 0.87912088 0.86813187
```